Modbus RTU Driver V7 — Library for Coolmay FX3G PLC

Abstract

The Modbus RTU Driver enables a Coolmay FX3G PLC to operate as a Modbus RTU master or slave on its two auxiliary RS485 ports (port 2 and port 3). On L02-series PLCs the same channel model also drives the built-in Ethernet port as a Modbus TCP master (client) — a channel whose `iPort` is `MB_PORT_TCP` is handled with the same scheduling logic as an RS485 channel — and `MB_MASTER_INIT_TCP` / `MB_SLAVE_INIT_TCP` configure that port as a Modbus TCP master (client) or slave (server). Communication channels are configured through an application-declared array `MB_CHANNELS` of type `MB_REG_50`; a single function block, `MB_PROCESS_50`, cycles through that array and performs the channel transfers on a **50 ms** tick.

The `MB_CHANNELS` array and the `c_MB_CHANNELS_NUM` capacity constant are declared by the application in its own global label list, sized to the number of channels it actually uses; `MB_PROCESS_50` cycles through exactly that many channels. This keeps the automatically-assigned device usage proportional to the number of channels actually configured.

Prerequisites

1. The following libraries must be installed in the project before this driver:
   - `Utils.sul`
   - `TimeControl.sul`
2. The TimeControl library's `TCO_TICKER_50` ticker must be running. It advances at **50 ms** intervals and drives the scheduling of `MB_PROCESS_50`.
3. The application must declare the Modbus channel storage in its own global label list before using `MB_PROCESS_50`:
   - `c_MB_CHANNELS_NUM` — `VAR_GLOBAL_CONSTANT` (`INT`), the upper bound (last valid index) of the channel array.
   - `MB_CHANNELS` — `VAR_GLOBAL` (`ARRAY [0..c_MB_CHANNELS_NUM] OF MB_REG_50`), the channel array itself.

   > The literal array bound must match `c_MB_CHANNELS_NUM`. For three channels (indices `0`–`2`), declare `c_MB_CHANNELS_NUM := 2` and `MB_CHANNELS : ARRAY [0..2] OF MB_REG_50`.
4. Modbus TCP channels additionally require an L02-series PLC with the built-in Ethernet interface and the `Utils` library's `L02_SET_IP` function. See § Modbus TCP (Ethernet).
5. This library is compatible with **GX Works 2 v1.631H** and later. Updates are available from the `coolmay/soft` directory.

Terminology

- **Channel** — A single configured Modbus read/write relationship, described by one element of `MB_CHANNELS`.
- **Register channel** — A channel that transfers 16-bit register data (function codes `H3`, `H4`, `H6`, `H10`). Its value buffer lives in the `D` device area.
- **Coil channel** — A channel that transfers discrete bit data (function codes `H1`, `H2`, `H5`, `HF`). Its value buffer lives in the `M` device area.
- **Scratch buffer** — A `D`-area workspace used by `MB_PROCESS_50` during a transfer, starting at the `mb_iBuffer` base device.

Type names: this document uses the IEC type names written in the GX Works 2 label CSV files. The Label Editor displays them as follows:

IEC type	GX Works 2 Label Editor display
`BOOL`	Bit
`INT`	Word [Signed]
`WORD`	Word [Unsigned] / Bit String [16-bit]
`DWORD`	Double Word [Unsigned]

Architectural Description

This library enables a Coolmay FX3G PLC to operate as a **Modbus Slave** or a **Modbus Master** on the secondary and tertiary RS485 ports (ports 2 and 3, respectively), and — on L02-series PLCs — as a **Modbus TCP Master** over the built-in Ethernet port. It provides a low-overhead interface for configuring and managing Modbus communication channels.

The Modbus channels are described in a user-declared global array `MB_CHANNELS` of type `MB_REG_50`. The library does not fix the channel count: the application declares the array and the `c_MB_CHANNELS_NUM` upper bound in its global label list, and `MB_PROCESS_50` cycles through exactly that many channels.

Coolmay PLC/HMI integrated units are equipped with two RS485 ports: port 2 is exposed on the terminal connector, and port 3 is exposed on the DB9 connector. L02-series PLCs also provide two RS485 ports, both on terminal connectors.

The recommended workflow is:

1. At startup (under the `M8002` initialisation pulse), configure the `MB_CHANNELS` array.
2. Initialise each RS485 port that is used with `MB_PORT_SETTINGS` and the relevant `MB_*_INIT_*` function. Re-trigger the initialisation periodically after startup (see the timing note in § Modbus Master). For a `MB_PORT_TCP` channel the Ethernet port is configured instead, as described in § Modbus TCP (Ethernet): `MB_MASTER_INIT_TCP` sets the role, port and server count and `L02_SET_IP` writes the addresses. To make the PLC a Modbus TCP server, call `MB_SLAVE_INIT_TCP` — that role needs no channel and no `MB_PROCESS_50` call.
3. Call `fbMbProcess` every scan (or at least faster than the shortest `tCycle`).

> **Calling convention:** the port-initialisation and settings POUs are *functions* that return their result through the function name. In GX Works 2 a function call used as a statement requires a left-hand side,

> so callers assign the result to a dummy bit, e.g. `M0 := MB_MASTER_INIT_PORT2(TRUE, PortSettings);`. The result value is not used elsewhere.

Function Blocks and Functions

Name	Type	Description
`MB_PORT_SETTINGS`	Function	Build the serial-port bit-field (parity, stop bits, baud rate).
`MB_MASTER_INIT_PORT2`	Function	Initialise the Modbus master on port 2.
`MB_MASTER_INIT_PORT3`	Function	Initialise the Modbus master on port 3.
`MB_SLAVE_INIT_PORT2`	Function	Initialise the Modbus slave on port 2.
`MB_SLAVE_INIT_PORT3`	Function	Initialise the Modbus slave on port 3.
`MB_SLAVE_INIT_TCP`	Function	Enable Modbus TCP slave (server) mode on the L02 Ethernet port.
`MB_MASTER_INIT_TCP`	Function	Configure the L02 Ethernet port as Modbus TCP master (client).
`MTB_SLAVE_PORT2`	Function	Reconfigure port 2 for the Mitsubishi protocol.
`MTB_SLAVE_PORT3`	Function	Reconfigure port 3 for the Mitsubishi protocol.
`MB_PROCESS_50`	Function Block	Cycle through the channels and perform their read/write transfers.
`MB_REG_50`	Structure	Per-channel configuration (declared by the application as `MB_CHANNELS`).

Global Constants and Variables

Access Modes

Constant	Value	Description
`MB_READ_WRITE`	`0`	Read and write.
`MB_READ`	`1`	Read only.
`MB_WRITE`	`2`	Write only.

Clear-on-Start Modes

Constant	Value	Description
`MB_CLEAR_NONE`	`0`	Clear nothing (default).
`MB_CLEAR_ALL`	`1`	Clear both the value buffer and the change-tracking buffer.
`MB_CLEAR_BUFFER`	`2`	Clear the change-tracking buffer only.

Serial Port Settings

Constant	Value	Description
MB_DL_7	0	7 data bits.
MB_DL_8	1	8 data bits.
MB_PARITY_NONE	0	No parity.
MB_PARITY_ODD	1	Odd parity.
MB_PARITY_EVEN	2	Even parity.
MB_STOPBIT_1	0	1 stop bit.
MB_STOPBIT_2	1	2 stop bits.
MB_BPS_600	0	600 bps.
MB_BPS_1200	1	1200 bps.
MB_BPS_2400	2	2400 bps.
MB_BPS_4800	3	4800 bps.
MB_BPS_9600	4	9600 bps.
MB_BPS_19200	5	19200 bps.
MB_BPS_38400	6	38400 bps.
MB_BPS_57600	7	57600 bps.
MB_BPS_115200	8	115200 bps.

> MB_DL_7 and MB_DL_8 are provided for completeness but are not selectable through MB_PORT_SETTINGS, which always configures **8 data bits**.

Ports

Constant	Value	Physical port
MB_PORT_2	0	RS485 port 2 — terminal connector (A, B)
MB_PORT_3	1	RS485 port 3 — DB9 connector (A1, B1)
MB_PORT_CAN	2	CAN port (H, L)
MB_PORT_TCP	3	Ethernet TCP port (L02-series PLCs only; see § Modbus TCP (Ethernet))

MB_PORT_SETTINGS (Function)

Returns the port-configuration value passed to MB_MASTER_INIT_* / MB_SLAVE_INIT_* when initialising a port as either Master or Slave.

Variable	Scope	Type	Description
`Parity`	INPUT	`INT`	One of: `MB_PARITY_NONE`, `MB_PARITY_ODD`, `MB_PARITY_EVEN`.
`StopBit`	INPUT	`INT`	One of: `MB_STOPBIT_1`, `MB_STOPBIT_2`.
`Baudrate`	INPUT	`INT`	One of: `MB_BPS_600`, `MB_BPS_1200`, `MB_BPS_2400`, `MB_BPS_4800`, `MB_BPS_9600`, `MB_BPS_19200`, `MB_BPS_38400`, `MB_BPS_57600`, `MB_BPS_115200`.

> **Note:** the data length is fixed at **8 data bits** and is not a parameter of this function.

Declare a local variable of type `DWORD` (e.g. `PortSettings`), then invoke the function:

```
VAR
    PortSettings : DWORD;
END_VAR

PortSettings := MB_PORT_SETTINGS(MB_PARITY_NONE, MB_STOPBIT_1, MB_BPS_9600);
```

Modbus Slave

`MB_SLAVE_INIT_PORT2`, `MB_SLAVE_INIT_PORT3`

These functions initialise the Modbus slave on port 2 or port 3 of the PLC, respectively.

Variable	Scope	Type	Description
`xInit`	INPUT	`BOOL`	Initialisation command. The slave is (re-)initialised on every rising edge.
`iAddress`	INPUT	`INT`	Modbus network address assigned to this PLC.
`PortSettings`	INPUT	`DWORD`	Return value of the `MB_PORT_SETTINGS` function.

> **Important — initialisation timing:** the function acts only on a rising edge of `xInit` (see § Modbus Master). Triggering it solely from the `M8002` first-scan pulse may not retain the port settings — feed the one-second clock `M8013` into `xInit` to re-apply them every second.

Example

The following snippet configures a slave with address `1` on port 2:

```
PortSettings := MB_PORT_SETTINGS(MB_PARITY_NONE, MB_STOPBIT_1, MB_BPS_9600);
M0 := MB_SLAVE_INIT_PORT2(M8013, 1, PortSettings);
```

No further configuration is necessary for the PLC to function as a slave on port 2. The same pattern applies to port 3. The variable `M0` is not referenced elsewhere in the program — it exists solely to satisfy the syntactic requirement that a function call used as a statement must have a left-hand side assignment.

> **Modbus TCP slave:** on an L02-series PLC the Ethernet port can serve as a Modbus TCP slave (server) as well — use `MB_SLAVE_INIT_TCP` instead, see § Modbus TCP (Ethernet) → § PLC as a Modbus TCP Server.

Modbus Master

`MB_MASTER_INIT_PORT2`, `MB_MASTER_INIT_PORT3`

These functions initialise the Modbus Master on port 2 (terminal connector) or port 3 (DB9 connector), respectively.

Variable	Scope	Type	Description
`xInit`	INPUT	`BOOL`	Initialisation command. The master is (re-)initialised on every rising edge.
`PortSettings`	INPUT	`DWORD`	Return value of the `MB_PORT_SETTINGS` function.

> **Important — initialisation timing:** the function acts only on a rising edge of `xInit`. Triggering it solely from the `M8002` first-scan pulse may not retain the port settings; feed the one-second clock `M8013` into `xInit` to re-apply them every second.

Example

```
PortSettings := MB_PORT_SETTINGS(MB_PARITY_NONE, MB_STOPBIT_1, MB_BPS_9600);
(* M8013 = 1 s clock: re-applies the port settings every second. *)
M0 := MB_MASTER_INIT_PORT2(M8013, PortSettings);
```

Port Reconfiguration (Mitsubishi Protocol)

`MTB_SLAVE_PORT2`, `MTB_SLAVE_PORT3`

These functions reconfigure port 2 or port 3 to the Mitsubishi proprietary protocol (not Modbus RTU), set the station address to `1`, and set the serial format to **38400 baud, 7 data bits, even parity, 1 stop bit**. They are used to switch a port back from Modbus RTU to the Mitsubishi protocol.

Variable	Scope	Type	Description
`xInit`	INPUT	`BOOL`	Reconfiguration command. The port is reconfigured on every rising edge.

Example

```
M0 := MTB_SLAVE_PORT2(TRUE);
```

`MB_PROCESS_50` (Function Block)

This function block orchestrates all read and write operations across the configured channels. It must be called every scan (or at least faster than the shortest channel `tCycle`).

> **Prerequisite:** The TimeControl library must be installed and `TCO_TICKER_50` must be running. This ticker advances at 50 ms intervals and drives the scheduling of channel operations.

Variable	Scope	Type	Description
`mb_xEnable`	INPUT	`BOOL`	Enables channel processing. When `FALSE`, processing is suspended; re-enabling restarts the channels from the beginning.
`mb_iBuffer`	INPUT	`INT`	Base `D` device number of the scratch buffer used during a transfer. The scratch buffer holds up to `iNum` words for register channels, or `⌈iNum / 16⌉` words for coil channels (one bit per coil). It must not overlap any channel's value or change-tracking buffer.
`mb_iClearOnStart`	INPUT	`INT`	Startup buffer-clearing mode: `MB_CLEAR_ALL`, `MB_CLEAR_BUFFER`, or `MB_CLEAR_NONE` (default).
`mb_iTimeoutCount`	INPUT	`INT`	Consecutive timeouts before a channel is suspended. Default (when `0`): `2`.
`mb_iSuspendRetry`	INPUT	`INT`	Suspended-channel retry interval, in 50 ms units. Default (when `0`): `80` (4 s).
`mb_iTimeoutTime`	INPUT	`INT`	Timeout duration, in 50 ms units. Default (when `0`): `4` (200 ms).
`mb_tStartDelay`	INPUT	`TIME`	Startup delay.
`mb_iTimeout`	OUTPUT	`INT`	Channel number (index) of the channel that timed out. Held for one scan; `-1` when no channel has timed out.

Timeout Tuning

> **Call-interval consideration.** The timeout is only evaluated when `MB_PROCESS_50` is executed, so the *effective* resolution of `mb_iTimeoutTime` is the block's actual call interval — not the 50 ms the name implies. `mb_iTimeoutTime` remains specified in 50 ms units (`value × 50 ms`). Set it so the timeout lands at least one full call period past the expected request completion. Example: block called every 100 ms and a request expected to finish in 2 calls (200 ms) → use `mb_iTimeoutTime := 6` (300 ms), not `4` (200 ms); otherwise the completion and the timeout are sampled on the same boundary and the request can be falsely marked as timed out.

Clear-on-Start

`mb_iClearOnStart` controls what `MB_PROCESS_50` does with each channel's two buffers when it initialises the channel array. This happens once, after the startup delay and before the first channel is processed. Every channel owns a **value buffer** (the actual register/coil values) and a **change-tracking buffer** (an internal copy used to detect local changes). The scheduler treats a value as "changed" — and, for a write-capable channel, writes it to the slave — whenever the value buffer differs from the change-tracking buffer.

Constant	Value	What is cleared
`MB_CLEAR_NONE`	`0`	Nothing (default).
`MB_CLEAR_ALL`	`1`	Value buffer and change-tracking buffer.
`MB_CLEAR_BUFFER`	`2`	Change-tracking buffer only.

What happens in each case

- `MB_CLEAR_ALL` **— read first.** Both buffers are zeroed, so they are equal and no local change is detected. Write-capable channels therefore issue **no write** on startup; read-capable channels simply perform their normal **read first** when the cycle elapses, filling the value buffer from the slave. The slave is treated as the source of truth.
- `MB_CLEAR_BUFFER` **— write current values.** The value buffer is preserved while the change-tracking buffer is zeroed. Every retained **non-zero** value therefore differs from its cleared change-tracking entry and is seen as "changed", so write-capable channels (`MB_READ_WRITE` or `MB_WRITE`) attempt to **write their current values to the slave** at the next write opportunity. A retained value of `0` matches the cleared entry and is **not** written.
- `MB_CLEAR_NONE` **— no side effects.** Nothing is cleared. Both buffers keep their contents, so no artificial "changed" state is introduced and no extra read or write is forced at startup. Retained values are written only if they genuinely differ from the retained change-tracking buffer.

Use cases

- `MB_CLEAR_NONE` — the PLC's `D`/`M` values are latched across restarts and already consistent with the change-tracking buffer. The driver is left untouched and performs only its normal `tCycle` reads/writes, with no startup write.
- `MB_CLEAR_ALL` — cold start or commissioning, or any situation where the slave is the source of truth and the driver should pull current slave values first. Because both buffers are cleared, it produces no spurious startup writes.
- `MB_CLEAR_BUFFER` — the PLC's retained value buffer is the source of truth and the slave may have lost its state (for example, the slave was power-cycled independently or replaced). On startup the driver re-pushes the retained values to the slave, re-synchronising the slave to the PLC.

> **Note:** The write-on-startup effect of `MB_CLEAR_BUFFER` applies to **non-zero** retained values only. Because the change-tracking buffer is zeroed, a retained `0` is indistinguishable from "unchanged" and is not written. If the retained value buffer is entirely zero, `MB_CLEAR_BUFFER` behaves like `MB_CLEAR_NONE` and no write occurs.

The `MB_REG_50` Structure

The application declares a global array `MB_CHANNELS` of type `MB_REG_50`, together with a global constant `c_MB_CHANNELS_NUM` that holds the array's upper bound (last valid index). The library references both labels directly — it does not declare them. `MB_PROCESS_50` cycles through channels `0` to `c_MB_CHANNELS_NUM`, so the number of channels is fully under application control (e.g. `c_MB_CHANNELS_NUM := 2` for 3 channels, `:=`

29 for 30 channels). Each channel may read or write up to **125** registers. The array must be configured once, typically on PLC startup under the M8002 initialisation pulse flag.

The structure contains the following user-configurable fields.

Variable	Type	Description
xEnabled	BOOL	Enable this channel for processing. When FALSE, the channel is skipped.
iDDevNum	INT	Base device number of the value buffer. Register channels use D devices; coil channels use M devices. Each channel consumes **2 × iNum** devices. See § iDDevNum for allocation rules.
iNum	INT	Number of registers or coils to read/write (max 125). Default (when 0): 1.
iReg	WORD	Starting Modbus register/coil address. Declared WORD (unsigned) so addresses above 32,000 can be specified.
iRF	WORD	Modbus read function code (H1–H4). Default (when 0): H3.
iWF	WORD	Modbus write function code (H5, H6, HF, H10). Default (when 0): H6.
iDev	WORD	Modbus slave device address. On a MB_PORT_TCP channel it is the remote server index (1–4); see § Modbus TCP (Ethernet).
tCycle	INT	Cycle interval for automatic reads/writes, in units of 50 ms. E.g. 20 = 1 second. Set to 0 for manual-only operation via xReadOnce / xWriteOnce.
iWR	INT	Read/write mode: MB_READ_WRITE, MB_READ, or MB_WRITE. Writes (including xWriteOnce / xWriteOnChange) require MB_READ_WRITE or MB_WRITE.
iPort	INT	Communication port identifier: MB_PORT_2, MB_PORT_3, MB_PORT_CAN, or MB_PORT_TCP. See § Ports.
xWriteOnChange	BOOL	When TRUE, changed values are written to the slave immediately upon detection. When FALSE, writes occur only when the tCycle interval elapses.
xReadOnce	BOOL	On a rising edge (FALSE → TRUE), triggers a single read of this channel. For manual-only reads, set tCycle to 0.
xWriteOnce	BOOL	On a rising edge (FALSE → TRUE), triggers a single write of this channel. For manual-only writes, set tCycle to 0.
xDone	BOOL	Read-only completion pulse. Set TRUE on completion of one read/write cycle; cleared at the start of the next cycle. Primarily useful for channels operating in manual mode (tCycle = 0).

Supported Read/Write Functions

Code	Mnemonic
H1	Read Coils
H2	Read Discrete Inputs

Code	Mnemonic
`H3`	Read Holding Registers
`H4`	Read Input Registers
`H5`	Write Single Coil
`H6`	Write Single Register
`HF`	Write Multiple Coils
`H10`	Write Multiple Registers

`iDDevNum` — Buffer Allocation

Each channel reserves **twice** the number of devices specified by `iNum`. Half of the allocation holds the actual register/coil values; the other half is used internally for change tracking. Register channels use the `D` device area; coil channels use the `M` device area.

If `iNum = 1` and `iDDevNum = 200`:

- `D200` stores the value.
- `D201` is reserved for internal use.

If `iNum = 5`:

- `D200`–`D204` store the values.
- `D205`–`D209` are reserved for internal use.

> **Caution:** Failure to account for the internal buffer when assigning `iDDevNum` values across consecutive channels will result in data corruption.

Consider the following **incorrect** configuration:

```
MB_CHANNELS[0].iDDevNum := 600;
MB_CHANNELS[0].iNum := 3;

MB_CHANNELS[1].iDDevNum := 603;
MB_CHANNELS[1].iNum := 1;
```

Channel 0 consumes devices `D600` through `D605` (3 values + 3 internal). The first free device is therefore `D606` — not `D603`. Assigning `iDDevNum := 603` to channel 1 causes its buffer to overlap with the internal area of channel 0, producing undefined behaviour.

The **correct** allocation is:

```
MB_CHANNELS[0].iDDevNum := 600;
MB_CHANNELS[0].iNum := 3;
```

```
MB_CHANNELS[1].iDDevNum := 606;
MB_CHANNELS[1].iNum := 1;
```

iWF — Write Function Selection

Two write-function codes are available for register channels:

- H6 **(Write Single Register):** When assigned to a multi-register channel, only the register whose value changed is written in a dedicated request.
- H10 **(Write Multiple Registers):** When assigned to a multi-register channel, a change to any register within the group triggers a single request that updates the entire group.

Guidance:

- For double-word values spanning multiple registers → use H10.
- For multiple independent single-register variables on the same channel → use H6.
- For a single-register channel → use H6.

The same principle applies to coil channels: use H5 (Write Single Coil), even when the channel spans multiple coils.

Complete Example

Before configuring channels, declare the storage in the project's global label list. The following declares three channels (indices 0–2):

Label	Class	Type / Value
c_MB_CHANNELS_NUM	VAR_GLOBAL_CONSTANT	INT = 2
MB_CHANNELS	VAR_GLOBAL	ARRAY [0..2] OF MB_REG_50

Declare the function-block instance and the port bit-field in the local label section of the POU:

```
VAR
    fbMbProcess  : MB_PROCESS_50;
    PortSettings : DWORD;
END_VAR
```

> M0, M1, M2, and M8002 in the example are device labels (auxiliary relays and the initialisation-pulse special relay) and do not require a VAR declaration.

```
IF M8002 THEN

    (* Timeout and clear-on-start tuning are supplied to fbMbProcess below. *)
    PortSettings := MB_PORT_SETTINGS(MB_PARITY_NONE, MB_STOPBIT_1, MB_BPS_9600);

    (* Automatic read/write channel on port 2 *)
    MB_CHANNELS[0].xEnabled := TRUE;
    MB_CHANNELS[0].iDDevNum := 600;
```

```
    MB_CHANNELS[0].iNum := 3;
    MB_CHANNELS[0].iReg := K16384;
    MB_CHANNELS[0].iRF := H3;
    MB_CHANNELS[0].iWF := H6;
    MB_CHANNELS[0].iDev := H1;
    MB_CHANNELS[0].tCycle := 20; (* 20 × 50 ms = 1 000 ms = 1 second *)
    MB_CHANNELS[0].xWriteOnChange := TRUE;
    MB_CHANNELS[0].iWR := MB_READ_WRITE;
    MB_CHANNELS[0].iPort := MB_PORT_2;

    (* Minimal setup — connects to a device via Port 3 *)
    MB_CHANNELS[1].xEnabled := TRUE;
    MB_CHANNELS[1].iDDevNum := 610;
    MB_CHANNELS[1].iReg := K16385;
    MB_CHANNELS[1].iDev := K16;
    MB_CHANNELS[1].tCycle := 4;
    MB_CHANNELS[1].iWR := MB_READ;
    MB_CHANNELS[1].iPort := MB_PORT_3;

    (* Manual-cycle read/write of a 3-register group *)
    MB_CHANNELS[2].xEnabled := TRUE;
    MB_CHANNELS[2].iNum := 3;
    MB_CHANNELS[2].iDev := H1;
    MB_CHANNELS[2].iPort := MB_PORT_3;
    MB_CHANNELS[2].iDDevNum := 10;
    MB_CHANNELS[2].iReg := H1000;
    MB_CHANNELS[2].iRF := H3;
    MB_CHANNELS[2].iWR := MB_READ_WRITE;
    MB_CHANNELS[2].tCycle := 0;
END_IF;

(* Multi-master mode is available on both ports for L02-series PLCs.
   M8013 = 1 s clock re-applies the port settings every second. *)
M0 := MB_MASTER_INIT_PORT2(M8013, PortSettings);
M0 := MB_MASTER_INIT_PORT3(M8013, PortSettings);

fbMbProcess(mb_xEnable := TRUE, mb_iBuffer := 100,
    mb_iClearOnStart := MB_CLEAR_ALL,
    mb_iTimeoutCount := 2,
    mb_iSuspendRetry := 80,
    mb_iTimeoutTime := 4);

(* Manually read channel 2 once *)
IF M1 THEN
    MB_CHANNELS[2].xReadOnce := TRUE;
    IF MB_CHANNELS[2].xDone = TRUE THEN
        M1 := FALSE;
        MB_CHANNELS[2].xReadOnce := FALSE;
    END_IF;
END_IF;

(* Manually write channel 2 once *)
IF M2 THEN
    D10 := 100;
```

```
    D11 := 100;
    D12 := 100;
    MB_CHANNELS[2].xWriteOnce := TRUE;
    IF MB_CHANNELS[2].xDone = TRUE THEN
        M2 := FALSE;
        MB_CHANNELS[2].xWriteOnce := FALSE;
    END_IF;
END_IF;
```

Post-configuration behaviour:

- `D600`, `D601`, and `D602` hold the three register values read from device address `1`, refreshed every second. If any value changes locally, the slave is updated immediately (`xWriteOnChange := TRUE`).
- `D610` holds the single register value read from device address `16`. Since this channel is configured as read-only (`MB_READ`), any local modification to `D610` is overwritten by the next read cycle (every 200 ms).
- Channel 2 (`tCycle = 0`) performs no automatic transfers. It is read or written only when `xReadOnce` / `xWriteOnce` is pulsed.

Alternative Write Confirmation

The expression `IF D10 = D13 AND D11 = D14 AND D12 = D15 THEN` implements a buffer-comparison strategy for verifying write completion.

As described in § `iDDevNum`, each channel reserves twice the number of devices specified by `iNum`. With `iDDevNum := 10` and `iNum := 3`:

- `D10`–`D12` hold the actual register values.
- `D13`–`D15` serve as the change-tracking buffer.

The change-tracking registers are updated together with the write. Equality between the value registers and their corresponding change-tracking registers therefore confirms that the write has completed.

```
(* Alternative write-confirmation technique *)
IF M2 THEN
    D10 := 200;
    D11 := 200;
    D12 := 200;
    MB_CHANNELS[2].xWriteOnce := TRUE;
    IF D10 = D13 AND D11 = D14 AND D12 = D15 THEN
        M2 := FALSE;
        MB_CHANNELS[2].xWriteOnce := FALSE;
    END_IF;
END_IF;
```

This technique is particularly useful when employing `xWriteOnChange` (rather than `xWriteOnce`) with a multi-register channel and write function `H6`. If, for example, two registers within a ten-register channel are modified, `H6` writes each one individually, setting `xDone` once per successful write. Counting two distinct `xDone` pulses can rapidly become unwieldy. In contrast, a single comparison of the form `IF D10 = D13 AND D11 = D14 AND …` confirms that all changed registers have been successfully propagated to the slave in a single check.

Modbus TCP (Ethernet)

On L02-series PLCs a channel can use the built-in Ethernet port instead of an RS485 port. A TCP channel is configured and used like any other channel — the same `MB_REG_50` fields, the same `tCycle`, `xReadOnce` / `xWriteOnce`, change-tracking, timeout and suspension rules — with two differences:

1. `iPort` must be set to `MB_PORT_TCP`.
2. The Ethernet interface and the remote servers are configured with the L02 network relays and registers — through the `MB_MASTER_INIT_TCP` / `MB_SLAVE_INIT_TCP` functions for the role, port and server count, and through the `L02_SET_IP` function of the `Utils` library for the IP addresses. `MB_PORT_SETTINGS` and the `MB_MASTER_INIT_PORT2` / `MB_MASTER_INIT_PORT3` functions are not used for a TCP channel: the Ethernet port has no serial settings.

RS485 channels and TCP channels can be mixed in the same `MB_CHANNELS` array.

> **Requirement:** an L02-series PLC with the built-in Ethernet interface. RS485-only Coolmay models have no usable `MB_PORT_TCP` channel.

Setup

Device	Value	Purpose
`D8395`	`4`	Modbus TCP master (client). `3` selects Modbus TCP server (see below). Written by `MB_MASTER_INIT_TCP` / `MB_SLAVE_INIT_TCP`.
`D8325`	`1`–`4`	Number of remote servers polled by this PLC. Written by `MB_MASTER_INIT_TCP`.
`R23812`	`502`	Modbus TCP port. Written by `MB_MASTER_INIT_TCP` / `MB_SLAVE_INIT_TCP`.
`M8197`	—	Apply the addresses and restart the interface. Set by `L02_SET_IP` on every write; the PLC resets it — no application code needed.
`M8193`	—	Network ready (`1` when the network is up).
`M8395`	—	Connection status: `1` = communication normal, `0` = abnormal.
`M8062`	—	Modbus timeout.
`M8063`	—	IP address conflict.
`R23815`, `R23816`	—	Number of Modbus packets sent / received.

IP addresses are written with `L02_SET_IP` from the `Utils` library (see the Utils manual). It takes a setting type followed by the four octets:

Setting type	Meaning
`IP_PLC_IP`	IP address of the PLC
`IP_PLC_MASK`	Subnet mask
`IP_PLC_GATEWAY`	Gateway

Setting type	Meaning
`IP_REMOTE1` … `IP_REMOTE4`	Remote server 1 … 4

`L02_SET_IP` sets `M8197` itself on every write and the PLC clears it again, so the application never writes `M8197` and needs no separate "apply" pulse. The function writes while its `xInit` input is `TRUE`, so pass a trigger bit that is `TRUE` for **one scan only** — `M8002` for the startup write, `MEP` of an HMI "save" pulse for a later change — and give all settings of one configuration the same trigger, so they are written and applied in a single scan:

```
xApplyIp := M8002 OR MEP(xSaveIp);

M0 := L02_SET_IP(xApplyIp, IP_PLC_IP, 192, 168, 1, 3);      (* PLC IP       *)
M0 := L02_SET_IP(xApplyIp, IP_PLC_MASK, 255, 255, 255, 0);  (* subnet mask  *)
M0 := L02_SET_IP(xApplyIp, IP_PLC_GATEWAY, 192, 168, 1, 1); (* gateway      *)
```

> Never tie `xInit` to `TRUE`: every scan is then a new write, so the interface would restart continuously and the network would never come up.

Remote Servers and `iDev`

For a TCP channel the `iDev` field selects the remote server: `iDev := 1` addresses the server configured with `IP_REMOTE1`, `iDev := 2` the server configured with `IP_REMOTE2`, and so on up to `iDev := 4`. At most four servers are supported, and `D8325` must match the number actually used. `iNum` remains the number of registers or coils transferred and is unrelated to the server selection.

Timeout

The PLC firmware retries a failed Modbus TCP request before reporting an error, so `mb_iTimeoutTime` must cover those retries: `mb_iTimeoutTime := 20` (1 s) is a practical starting value. Use a startup delay (`mb_tStartDelay := T#10s`) so that the first request waits for `M8193` (network ready).

Modbus TCP Master Configuration

`MB_MASTER_INIT_TCP` (Function)

Configures the L02 built-in Ethernet port as a Modbus TCP master (client): the Modbus TCP port, the number of remote servers polled by this PLC, and the master role.

Variable	Scope	Type	Description
`xInit`	INPUT	`BOOL`	Configuration trigger. The configuration is written while it is `TRUE`, so pass a one-scan pulse — `MEP(xApplyIp)` — as shown in the example.
`iTcpPort`	INPUT	`INT`	Modbus TCP port. A non-zero value is written to `R23812`; `0` keeps the current port (`502`).
`iServers`	INPUT	`INT`	Number of remote servers polled by this PLC, clamped to `1`–`4` and written to `D8325`.

The function writes `R23812`, `D8325` and `D8395`, but not `M8197` — the values are applied and the interface restarts on the `L02_SET_IP` call that sets `M8197`. Call `MB_MASTER_INIT_TCP` therefore **before** the `L02_SET_IP` writes in the same scan.

The body is deliberately **level**-triggered (`IF xInit THEN`, no internal `MEP`): the application owns the edge and passes a one-scan pulse, which keeps the trigger visible at the call site and lets the same pulse gate several calls. Never tie `xInit` to `TRUE`.

> - Do not write `D8395`, `D8325` or `R23812` from the application when this function is used.
> - `iServers` must match the number of `IP_REMOTE1` … `IP_REMOTE4` addresses written with `L02_SET_IP`; the `iDev` field of a TCP channel selects one of them.
> - Master mode is mutually exclusive with `MB_SLAVE_INIT_TCP`: one PLC cannot poll remote servers and serve requests as a slave over the same port at the same time. Select one role per project.

Example — Modbus TCP Master

Declare the function-block instance and the status labels in the local label section:

```
VAR
    fbMbProcess    : MB_PROCESS_50;
    xSaveIp        : BOOL; (* pulse to re-apply the IP addresses   *)
    xApplyIp       : BOOL; (* apply trigger passed to L02_SET_IP   *)
    xNetworkReady  : BOOL;
    xConnection    : BOOL;
    xModbusTimeout : BOOL;
    xIPConflict    : BOOL;
    iTX            : INT;
    iRX            : INT;
END_VAR
```

In the POU body, configure the Ethernet port, the channel and the scheduler:

```
(* --- Ethernet setup of the L02 built-in port ---------------------- *)
(* One trigger for the whole configuration: first scan, and again     *)
(* after an address or port change.                                   *)
xApplyIp := M8002 OR MEP(xSaveIp);

(* Role, port and server count first - applied by the writes below.   *)
M0 := MB_MASTER_INIT_TCP(xApplyIp, 502, 1); (* 502, one remote server *)

(* IP addresses - the last write sets M8197 and applies everything.   *)
M0 := L02_SET_IP(xApplyIp, IP_PLC_IP, 192, 168, 1, 3);      (* PLC IP       *)
M0 := L02_SET_IP(xApplyIp, IP_PLC_MASK, 255, 255, 255, 0);  (* subnet mask  *)
M0 := L02_SET_IP(xApplyIp, IP_PLC_GATEWAY, 192, 168, 1, 1); (* gateway      *)
M0 := L02_SET_IP(xApplyIp, IP_REMOTE1, 192, 168, 1, 5);     (* remote 1     *)

(* --- Status (monitoring) ------------------------------------------ *)
xNetworkReady  := M8193;        (* 1 = network chip ready   *)
xConnection    := M8395;        (* 1 = communication normal *)
```

```
xModbusTimeout := M8062;        (* 1 = Modbus timeout       *)
xIPConflict    := M8063;        (* 1 = IP address conflict  *)
iTX            := R23815;       (* packets sent             *)
iRX            := R23816;       (* packets received         *)

(* --- Modbus channel ------------------------------------------------ *)
IF M8002 THEN
    MB_CHANNELS[0].xEnabled       := TRUE;
    MB_CHANNELS[0].iDDevNum       := 1000;  (* values D1000, change tracking D1001
*)
    MB_CHANNELS[0].iNum           := 1;
    MB_CHANNELS[0].iReg           := K16384;
    MB_CHANNELS[0].iRF            := H3;    (* read holding registers  *)
    MB_CHANNELS[0].iWF            := H6;    (* write single register   *)
    MB_CHANNELS[0].iDev           := 1;     (* remote server 1         *)
    MB_CHANNELS[0].tCycle         := 20;    (* 20 x 50 ms = 1 s        *)
    MB_CHANNELS[0].iWR            := MB_READ_WRITE;
    MB_CHANNELS[0].iPort          := MB_PORT_TCP; (* mandatory          *)
    MB_CHANNELS[0].xWriteOnChange := TRUE;
END_IF;

(* --- Scheduler ----------------------------------------------------- *)
fbMbProcess(
    mb_xEnable       := TRUE,
    mb_iBuffer       := 900,
    mb_iClearOnStart := MB_CLEAR_NONE,
    mb_iTimeoutCount := 100,
    mb_iSuspendRetry := 4,
    mb_iTimeoutTime  := 20,
    mb_tStartDelay   := T#10s);
```

After the startup delay the scheduler reads `D1000` every second from Modbus register `16384` of the server at `192.168.1.5` (`iDev = 1`). A local change to `D1000` is written to the server immediately, because `xWriteOnChange := TRUE`.

PLC as a Modbus TCP Server

In the server (slave) role the PLC answers incoming Modbus TCP requests itself instead of polling remote servers. No `MB_CHANNELS` entry and no `MB_PROCESS_50` call are needed — the driver is not involved, and the only library call is `MB_SLAVE_INIT_TCP`.

`MB_SLAVE_INIT_TCP` (Function)

Enables Modbus TCP slave (server) mode on the L02 built-in Ethernet port.

Variable	Scope	Type	Description
`xInit`	INPUT	`BOOL`	Initialisation command. The TCP slave mode is enabled on every rising edge.
`iTcpPort`	INPUT	`INT`	Modbus TCP port. A non-zero value is written to `R23812`; `0` keeps the current port (`502`).

The function does not configure the network addresses — write them with `L02_SET_IP` from the `Utils` library (§ Setup above). `L02_SET_IP` sets `M8197` on every write, which applies both the addresses and the TCP port written by this function.

Example

```
(* Trigger: first scan, and again after an address or port change. *)
xApplyIp := M8002 OR MEP(xSaveIp);

M0 := L02_SET_IP(xApplyIp, IP_PLC_IP, 192, 168, 1, 100);     (* PLC IP      *)
M0 := L02_SET_IP(xApplyIp, IP_PLC_MASK, 255, 255, 255, 0);   (* subnet mask *)
M0 := L02_SET_IP(xApplyIp, IP_PLC_GATEWAY, 192, 168, 1, 1);  (* gateway     *)

(* Modbus TCP server on port 502. *)
M0 := MB_SLAVE_INIT_TCP(M8013, 502);
```

Feeding the one-second clock `M8013` into `xInit` re-applies the server mode every second, exactly as the RS485 initialisation functions are re-triggered after startup.

> - When this function is used, do not write `D8395` from the application — it owns that register.
> - Every `L02_SET_IP` write sets `M8197` and restarts the L02 interface, including the value written to `R23812`. Write the addresses first and call `MB_SLAVE_INIT_TCP` afterwards.
> - Server mode is mutually exclusive with the `MB_PORT_TCP` client channels: one PLC cannot poll remote servers and serve requests as a slave over the same port at the same time. Select one role per project.

Notes

> - `iPort := MB_PORT_TCP` is mandatory; if it is left at an RS485 port, the channel is driven on that serial port instead of the Ethernet port.
> - The IP addresses are written explicitly by the `L02_SET_IP` calls above; re-pulse `xSaveIp` after any address change so that the new addresses are applied.
> - `M8062` (Modbus timeout) and `M8395` (connection flag) are network indications that complement the driver's per-channel `mb_iTimeout` output. The suspend/retry policy is the same as for an RS485 channel.

Timeout and Suspension Mechanism

The library implements a channel-suspension policy for fault tolerance. If a channel fails to receive a response for `mb_iTimeoutCount` consecutive attempts, it is flagged as **suspended**. Once suspended, the channel is polled at a reduced rate — once every `mb_iSuspendRetry` interval. As soon as a valid response is received, the suspension flag is cleared and the channel resumes its normal cycle interval as defined by `MB_CHANNELS[*].tCycle`.

When a request times out, `MB_PROCESS_50.mb_iTimeout` reports the index of the failing channel (0-based) for one scan and is `-1` otherwise. This output can be used to react to a specific channel timing out, e.g. to latch an alarm or log the event.

A request that completes within its time window is always counted as a success, even if the completion arrives in the same block call as the timeout deadline. Set `mb_iTimeoutTime` comfortably longer than the expected request and response round-trip time and account for the block's call interval (see Timeout Tuning above); if it is too short, every request is abandoned as timed out, the channel never clears its timeout counter, and it enters the suspend/retry loop.